EXISTING COURSES:

0) Introduction to Basic Programming Concepts - Data Science Workshop
ID: DSW-000
About This Course

Target Group
Anybody did never attend a programming class or forgot almost everything…

Content
In this training we are explaining very basic programming skills. If you have never heard about if-statements, while loops or data structures / organization of data or don't know that computer programs need to define variables by content (text or numeric) and size you need to attend this class. All others can skip it, since we will explain in the other classes how to do it in Python (but not why it is needed).

- Type of variables – and why programs need to define them
- Why do we need to organize our data in different data structures?
- What are if-Statements
- How can while-loops be helpful

Prerequisites

- Laptop (if classroom training)
- EnergyAI Account (Go to: https://energyai.siemens.cloud/ (using Chrome, Edge or Firefox; Zscaler has to be active) and Request access to "Analyse" at least 1 week before the start of the training.)

1) Python & Numpy - Data Science Workshop
ID: DSW-001
About This Course

Target Group
Any engineer who needs to analyze data or wants to program python scripts.
Content

Content
Python is a powerful tool to analyze huge amounts of data with high calculation speed. This module gives you an overview of the different data structures in Python (lists, dictionaries, etc.). We will also introduce the algebra library Numpy, which is the backbone of Pythons big data options. You will learn basic commands such as loops, if statements and definitions. Our goal is to introduce you to some basic concepts. You will know enough to start yourself trial/error learning and consulting online sources.

- Intro Jupyter Notebook
- Lists, Object Localization
- Loops
- Dictionares
- If
- Definitions
- Intro to Numpy
- ND-Arrays

Prerequisites

- Have a very basic understanding of general concepts like loops, if-statements or the need to define data types/ or attend “0) Introduction to Basic Programming Concepts - Data Science Workshop, DSW-000”
- Laptop (if classroom training)
- EnergyAI Account (Go to: https://energyai.siemens.cloud/ (using Chrome, Edge or Firefox; Zscaler has to be active) and Request access to “Analyse” at least 1 week before the start of the training.)

2) Pandas - Data Science Workshop

Intern

ID: DSW-002

About This Course

Target Group

Any engineer who needs to analyze data or wants to program python scripts.

Content

Pandas is the part of Python used to process data. To make efficient calculations, we need to arrange data in statistical format (like for Excel Pivot-Tables or Minitab: columns for variables, rows for units, only one data type in each column). Also, we like to avoid missing values, correct data errors etc. All this data manipulation can be done very effectively with Pandas. After the training, you should be able to load, manipulate and analyze your own data in Python. Like always our goal is to introduce and practice some basic concepts. You will know enough to start your own learning by trial & error and consulting online sources.

- Intro pandas, dataframes
- Importing file
- Selection of objects
- Resample and Fill
- Calculation with datadrames
- Connecting dataframes
- Seaborn:
    - Overview
    - X-Axis = Continuous Data
    - X-Axis = Categorial Data

Prerequisites

- Attend “1) Python & Numpy - Data Science Workshop, DSW-001” or read through the contend on EnergyAI (shared > users > kay.toepfer@siemens.com >Data Science Workshop > Python.ipynb; Numpy.ipynb)
- Laptop (if classroom training)
- EnergyAI Account (Go to: https://energyai.siemens.cloud/ (using Chrome, Edge or Firefox; Zscaler has to be active) and Request access to “Analyse” at least 1 week before the start of the training.)

Introduction to Machine Learning 1 (for Everybody)
ID: MLW-001
About This Course

Target Group
Anybody who wants to understand what Machine Learning is about and how it could be useful in our business.

Content
Machine learning is one domain of artificial intelligence and the core of many impressive results Google, Facebook and others present to us every day. But what is Machine learning exactly? How does it work? And how can it actually help also our business? This training will give an introduction for beginners. On the end you will be familiar with some tools, there pros and cons of Machine Learning and will even have built a neuronal network yourself.

- What is Machine Learning?
- Machine Learning types
- Classification Algorithms
- Neuronal Network Regression
- Application of Machine Learning in our business
- Limits of Machine Learning (Bias and Overfitting)
- Pros and cons of machine learning

Prerequisites

- Laptop (if classroom training)
- EnergyAI Account (Go to: https://energyai.siemens.cloud/ (using Chrome, Edge or Firefox; Zscaler has to be active) and Request access to “Analyse” at least 1 week before the start of the training.)

Introduction to Machine Learning 2 (Adding Background)
ID: MLW-002
About This Course

Target Group
Employees who are planning to actually use ML for their work and therefore need to understand some background to apply it properly.

Content
In this training we will build on the training Machine Learning 1 and dig deeper. We will explain some critical mathematical background which will make it easier for you to interpret the results of decision tree, random forest and neuronal networks. The goal is, that you can build simple models yourself and to have an introduction to some basic concepts. You will know enough to start further exploring and learning.

- Mathematical background on Decision Tree analysis
- Mathematical background on Random Forrest analysis
- Mathematical background on Neuronal Network Regression
- Limits of Machine Learning (Bias and Overfitting)
- Difference Multiple Regression vs. Neuronal Network Regression
- Engineering application examples
-

Prerequisites

- Attending “Introduction to Machine Learning 1 (for Everybody), MLW-001” or read through slides
- Laptop (if classroom training)
- EnergyAI Account (Go to: https://energyai.siemens.cloud/ (using Chrome, Edge or Firefox; Zscaler has to be active) and Request access to “Analyse” at least 1 week before the start of the training.)